

XTRACT

WP5 | T5.1

Outdoor UAV configuration

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Drone provision and platform validation

Funded by the European Union

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T5.1 – Outdoor platform evolution

From completed assembly to field testing and the final mine demonstration

01

June 2026

Assembly & lab checks

- Frame and systems assembled
- Power and avionics integrated
- Lab and initial flight tests

02

September 2026

Field-tested platform

- Outdoor configuration finalised
- Multiple field-test campaigns
- Flights with multispectral camera

03

Final Björkdal pilot

Mine demonstration

- Outdoor flights performed
- Multispectral camera onboard
- Detailed findings: USAL / WP8

Outdoor UAV platform completed, field-tested and deployed in the final pilot.

T5.1 – Outdoor configuration completed

ACCELIGENCE provides the UAV platform for the joint sensing missions



Outdoor UAV assembly • previous GA

Structure & manufacturing

Mechanical layout completed; frame manufactured and assembled.

Integrated flight systems

Propulsion, power distribution, communications and flight control integrated.

Mission payload support

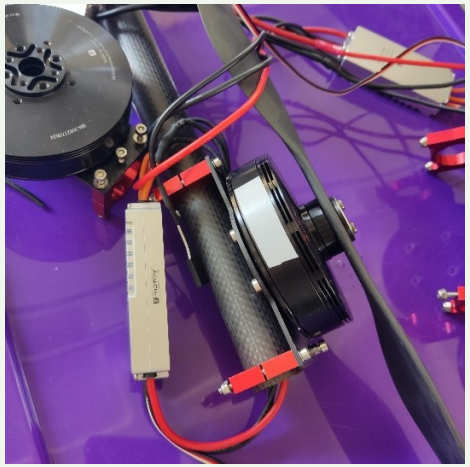
Platform prepared for payload integration; multispectral-equipped flights completed.

Completed contribution: outdoor UAV design, manufacturing, assembly and flight capability.

T5.1 – Platform testing and validation

Laboratory checks, progressive outdoor flights and mission-load simulation

Laboratory checks



Thrust, current and thermal behaviour checks.

Outdoor flight tests



Controller and propulsion tuning in outdoor flights.

Payload simulation



2.5 kg dummy payload for mission-load simulation.

The test programme supported platform readiness for the joint USAL missions.

T5.1 – Joint missions with USAL

Drone provision and flight capability support the sensor results reported by USAL

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UAV platform

- Outdoor configuration and assembly
- Drone provision and flight readiness
- Platform testing and payload support



Joint flight campaigns

USAL

Sensor results

- Coordination of sensing requirements
- Sensor data processing and analysis
- Interpretation and reporting of results

Björkdal: outdoor flights with the multispectral camera completed.

Detailed pilot scenarios: WP8. Sensor results and interpretation: USAL.

T5.1 – Risks and mitigations

Managed platform risks; pilot-specific lessons to be consolidated after the debrief

Risk / constraint	Potential impact	Mitigation implemented	Evidence / follow-up
Payload loading	Added mass affects flight behaviour	Thrust evaluation; outdoor flights with a 2.5 kg dummy payload.	Laboratory checks and mission-load simulation
Electrical / thermal behaviour	Excess current or overheating	Current and thermal monitoring during laboratory propulsion checks.	Lab test campaign completed
Payload integration	Interfaces or sensor readiness delay missions	Coordination with USAL; progressive integration and multispectral-equipped flights.	Multispectral flights completed; sensor results reported by USAL
Outdoor deployment	Field conditions affect platform operation	Progressive controller tuning and repeated outdoor flight campaigns.	Björkdal flights completed; detailed feedback pending

Add the confirmed Björkdal issues, fixes and residual limitations after the team debrief.

T5.1 – Next steps

Before Christmas | pilot feedback and joint reporting

01 Align the reporting with USAL

Connect ACCELIGENCE platform evidence with the sensor results and analysis reported by USAL.

02 Consolidate the pilot feedback

Record platform findings, observed issues and mitigations after the flight-team debrief.

03 Prepare the platform evidence

Compile configuration, testing and deployment evidence for WP5 reporting and the Final Review.

Outdoor platform completed; next steps concern feedback and reporting.

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T5.1 – Final Review preparation

Until the end of the project | platform evidence and partner alignment

Completed outdoor UAV

Record the final platform configuration, integration interfaces and operating setup.

Testing and deployment

Link laboratory checks, dummy-payload tests, field campaigns and the Björkdal flights.

Joint WP5 story with USAL

Connect drone provision to the sensor results presented by USAL; keep pilot detail in WP8.

ACCELIGENCE provides the drone; USAL presents the sensor results.

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Outdoor UAV: completed, tested and deployed

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https://www.linkedin.com/company/xtract-project/

https://www.youtube.com/channel/UC9eNoqHV2BLXc37Q_CTwFuQ

https://twitter.com/xtract_project

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